

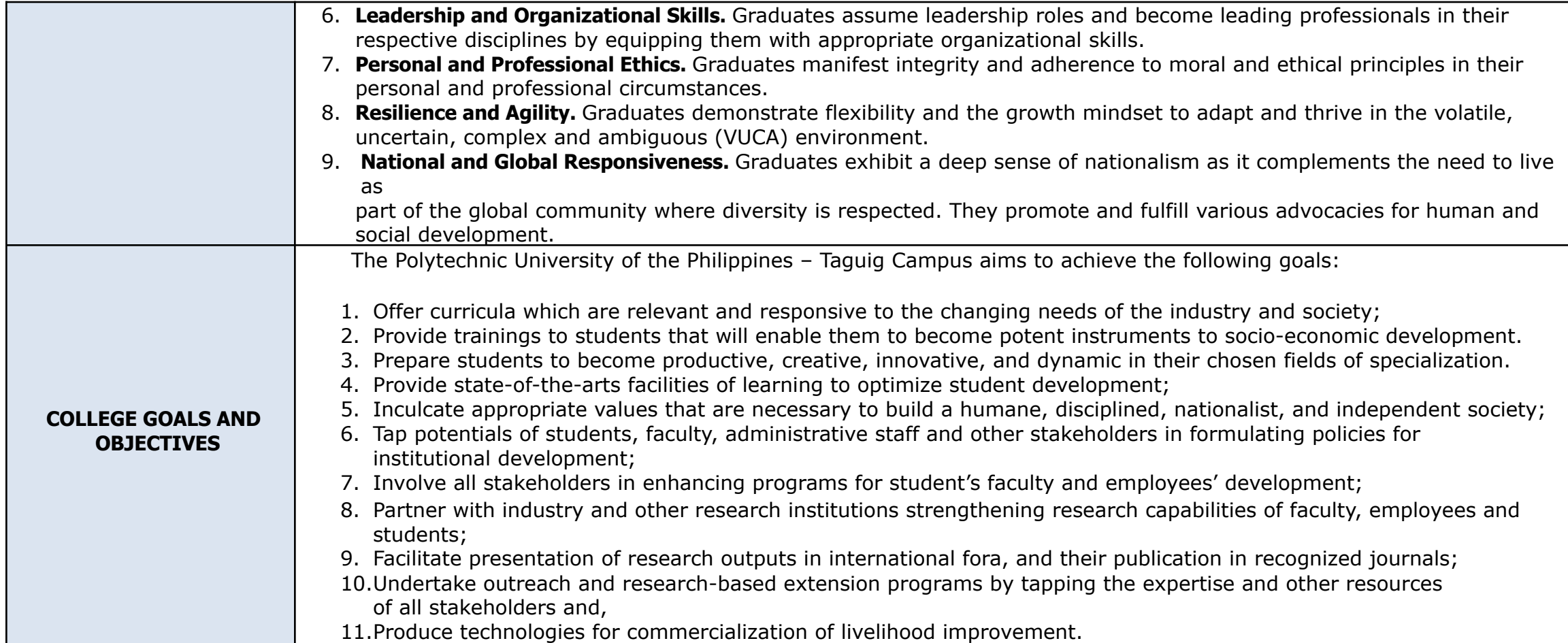


COURSE SYLLABUS						
Course Code	INTE 302	Course Title	Information Assurance and Security 1		Course Credit	3
Course Description		This course is designed to provide students with a foundational understanding of information assurance and security principles. The materials covered in this course include concepts such as information security fundamentals, risk management, security policies and procedures, cryptography basics, network and system security, access control, ethical and legal considerations, and current trends and issues in cybersecurity.				
Pre-Requisites		INTE 301		Co-Requisites	None	
VISION		A Leading Comprehensive Polytechnic University in Asia				
MISSION		Advance an inclusive, equitable, and globally relevant polytechnic education towards national development.				
QUALITY POLICY STATEMENT		The Polytechnic University of the Philippines commits to provide inclusive and equitable quality education and promote lifelong learning opportunities for human advancement and social transformation through re-engineered polytechnic academic programs. Toward this end, we, the members of the PUP Community will vigorously and steadfastly endeavor to continuously improve the standard of University services to the satisfaction of our clients through the adoption and continuous review of our Quality Management System.				



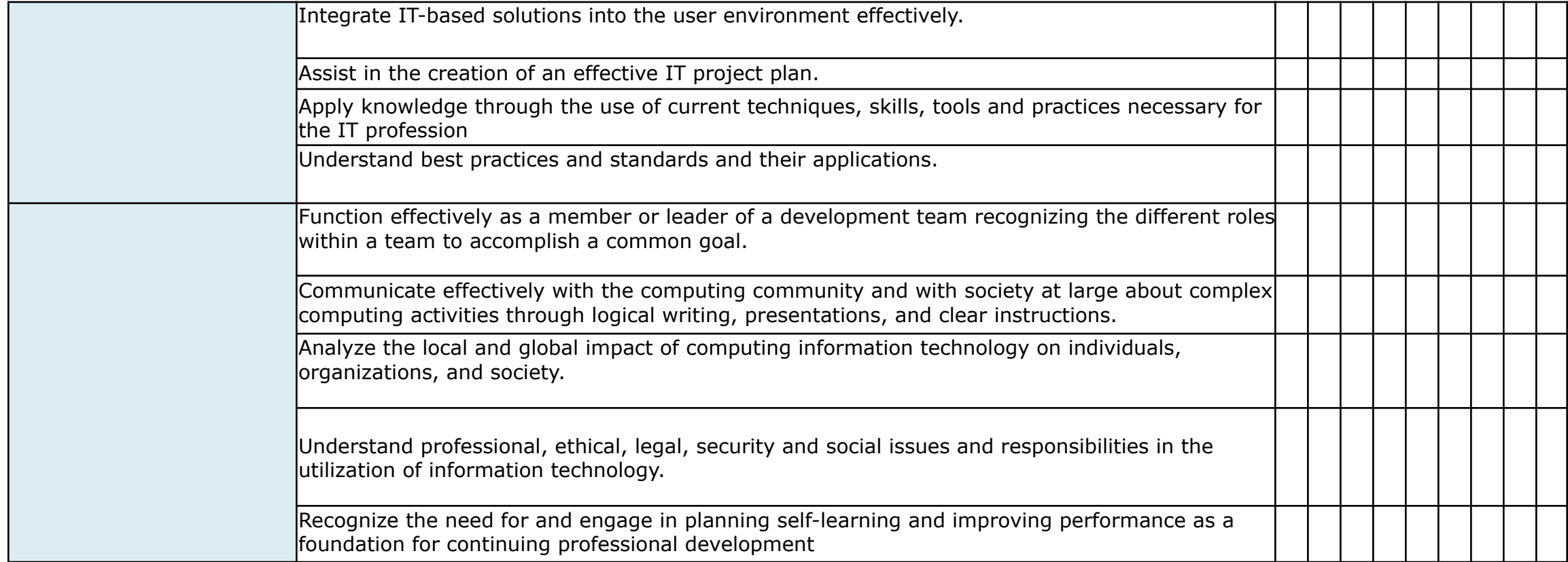
**INSTITUTIONAL
LEARNING OUTCOMES
(ILO)**

- As a polytechnic state university, PUP shall develop its students to possess:
1. **Critical and Creative Thinking.** Graduates use their rational and reflective thinking as well as innovative abilities to life situations in order to push boundaries, realize possibilities, and deepen their interdisciplinary, multidisciplinary, and/or transdisciplinary understanding of the world.
 2. **Effective Communication.** Graduates apply the four macro skills in communication (reading, writing, listening, and speaking), through conventional and digital means, and are able to use these skills in solving problems, making decisions, and articulating thoughts when engaging with people in various circumstances.
 3. **Strong Service Orientation.** Graduates exemplify strong commitment to service excellence for the people, the clientele, industry, and other sectors.
 4. **Adept and Responsible Use or Development of Technology.** Graduates demonstrate optimized and responsible use of state- of-the-art technologies of their profession. They possess digital learning abilities, including technical, numerical, and/or technopreneurial skills.
 5. **Passion for Lifelong Learning.** Graduates perform and function in society by taking responsibility in their quest for further improvement through lifelong learning.





PROGRAM LEARNING OUTCOMES (PLO)	Based on CMO	Alignment to ILOs								
		1	2	3	4	5	6	7	8	9
	Apply knowledge of computing, science, and mathematics appropriate to the discipline.									
	Analyze complex problems, and identify and define the computing requirements appropriate to its solution									
	Identify and analyze user needs and take them into account in the selection, creation, evaluation and administration of computer-based systems.									
	Design, implement, and evaluate computer-based systems, processes, components, or programs to meet desired needs and requirements under various constraints.									



COURSE LEARNING OUTCOMES (CLO)	Alignment to PLOs							
	1	2	3	4	5	6	7	8
Explain the fundamental concepts, principles, and objectives of Information Assurance and Security, including the CIA Triad (Confidentiality, Integrity, Availability).								

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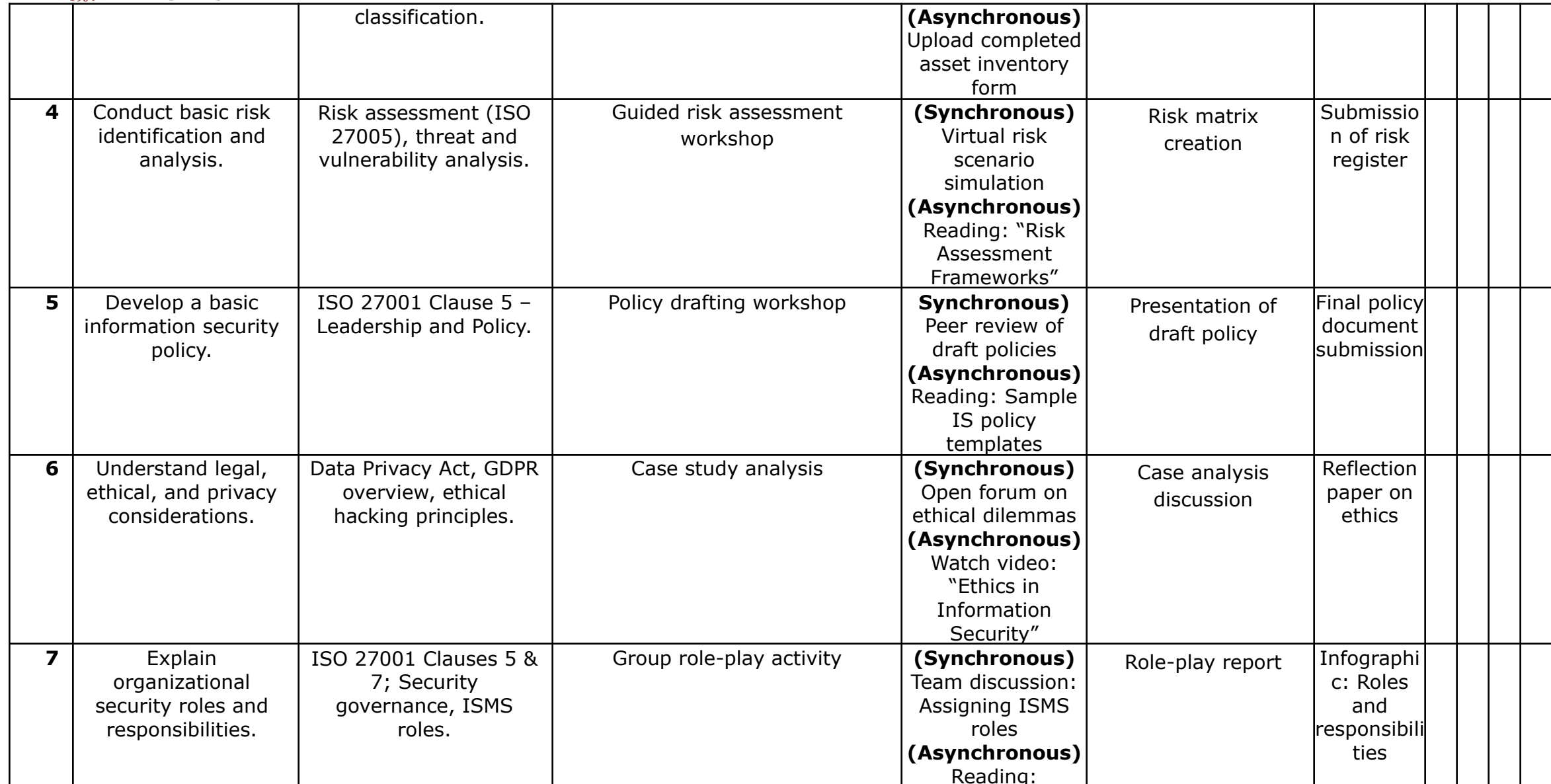


	Describe the structure, purpose, and operation of an Information Security Management System (ISMS) based on ISO 27001 standards.								
	Identify and classify information assets and evaluate their value, sensitivity, and required protection level.								
	Conduct basic risk identification, assessment, and treatment planning for information security threats and vulnerabilities.								
	Develop and apply organizational security policies and controls addressing physical, logical, and administrative security.								
	Demonstrate awareness of legal, ethical, and privacy issues, including compliance with the Data Privacy Act and related global regulations.								
	Implement basic incident response, business continuity, and disaster recovery procedures in alignment with organizational policies.								
	Integrate all learned principles and controls to propose an ISO 27001-aligned ISMS framework applicable to a simulated organization.								



OUTCOMES-BASED TEACHING AND LEARNING PLAN (OBTL PLAN)

Week	Desired Learning Outcomes (DLOs)	Learning Content/Topics	Instructional Delivery Design		Assessment		Alignment to CLOs			
			Face-to-Face	Flexible Teaching and Learning Activities (FLTAs)	Face-to-Face	FLTAs	1	2	3	4
1	Understand the fundamentals of Information Assurance and Security (IAS).	Overview of IAS; importance of confidentiality, integrity, availability (CIA Triad); Introduction to ISO 27001.	Lecture & discussion	(Synchronous) Orientation via video conference (Asynchronous) Reading: ISO 27001 overview PDF	Formative Assessment (Sharing of learning)	Online quiz on CIA Triad	X			
2	Explain the concept and scope of Information Security Management Systems (ISMS).	ISO 27001: Structure, purpose, and benefits of ISMS; PDCA cycle (Plan-Do-Check-Act).	Case study discussion	(Synchronous) Live walkthrough of ISO 27001 clauses (Asynchronous) Self-paced reading on ISMS components	Class discussion output	Short essay "Why ISMS Matters"				
3	Identify and classify types of information assets.	Asset management principles; ISO 27001 Clause 8.1; Asset inventory and	Workshop: Creating an asset inventory	(Synchronous) Demonstration of asset inventory tool	Group presentation of asset list	Asset register submission				



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				Governance framework PDF					
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2025

7th in the Philippines
92nd in Southeast Asia
541-560 in Asia



The WORLD
UNIVERSITY
RANKINGS
for INNOVATION

Rank 21
GENERATIVE AI
Rank 39
STUDENT SUPPORT
Rank 50
SYMBOL/



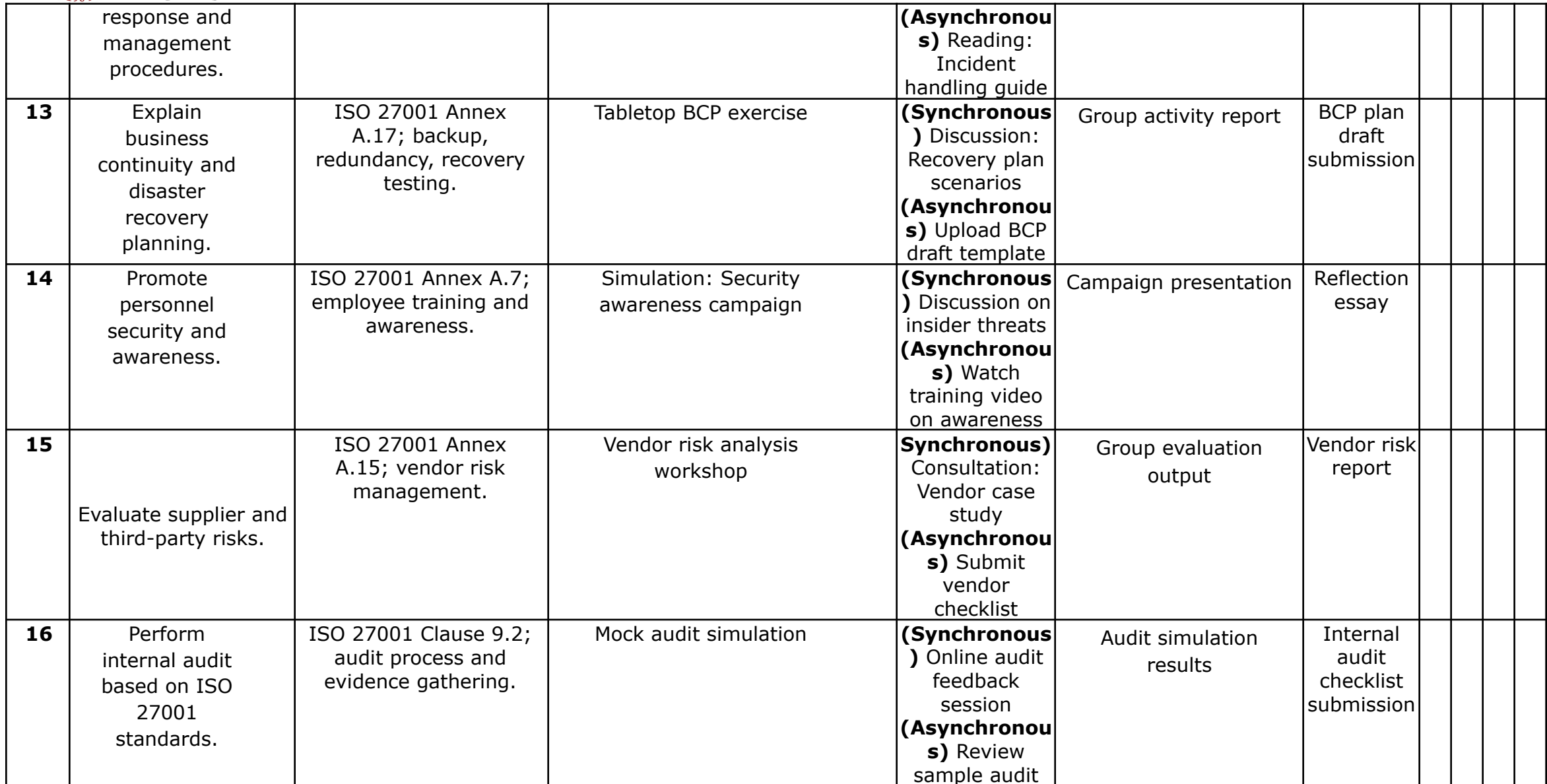
★★★★★
Employability
★★★★★
Teaching
★★★★★
Academic
Development
★★★★★
Social Responsibility
Specialist Criteria: Bachelor
of Science in Nutrition and
Dietetics (BSND)
★★★★★
Inclusiveness
★★★★★
Internationalization



DEBUTS AT 1501+

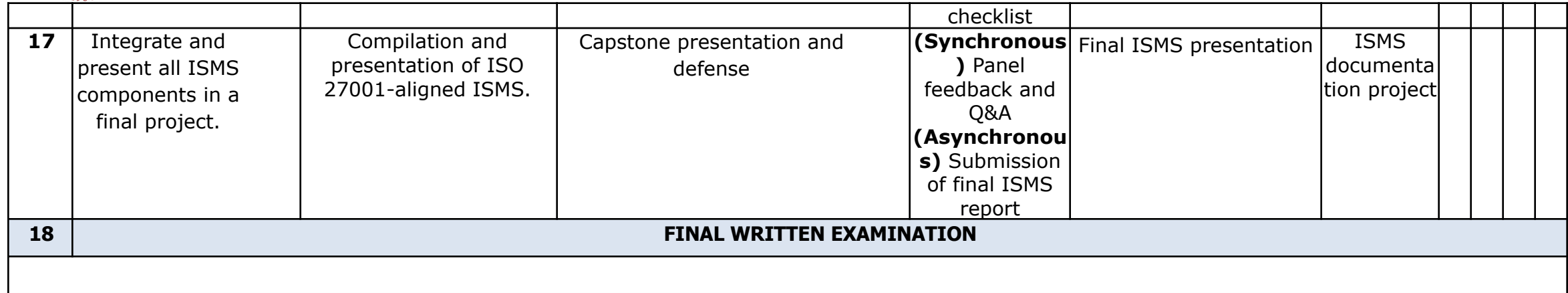


8	Recognize and apply physical, environmental, and logical access control mechanisms.	ISO 27001 Annex A.9 & A.11; facility access, authentication, and account management controls.	Combined lab mapping and hands-on activity	(Synchronous) Virtual walkthrough of secured facility and demo on access control setup (Asynchronous) Upload security layout map and screenshots of configuration	Group practical output (facility & access setup)	Security control checklist and brief reflection				
9	MIDTERM WRITTEN EXAMINATION									
10	Identify and mitigate network security threats.	ISO 27001 Annex A.13; network protection, IDS/IPS, firewalls.	Lab: Configure basic firewall and IDS	(Synchronous) Real-time troubleshooting (Asynchronous) Reading: Common network threats	Practical network exercise	Network security report				
11	Implement basic cryptographic controls.	ISO 27001 Annex A.10; encryption methods, PKI.	Lab activity: Encrypt/decrypt data files	(Synchronous) Q&A: Key management (Asynchronous) Upload screenshots of encryption process	Lab output	Quiz on cryptography concepts				
12	Demonstrate incident	ISO 27001 Annex A.12 & A.16; incident handling.	Simulated incident response drill	(Synchronous) Post-incident review meeting	Incident role-play report	Incident report submission				



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LIST OF REFERENCES

BOOKS:

WEBSITE:

International Organization for Standardization. (2022). *ISO/IEC 27001:2022 – Information security, cybersecurity and privacy protection - Information security management systems - Requirements*. ISO. <https://www.iso.org/standard/82875.html>

Shukla, A., Katt, B., Nweke, L. O., Yeng, P. K., & Weldehawaryat, G. K. (2021). *System security assurance: A systematic literature review*. arXiv. <https://arxiv.org/abs/2110.01904> arXiv

International Organization for Standardization. (2022). *ISO/IEC 27001:2022 – Information security, cybersecurity and privacy protection - Information security management systems - Requirements*. ISO. <https://www.iso.org/standard/27001>

GeeksforGeeks. (2022, April 26). *Information assurance vs information security*. GeeksforGeeks. <https://www.geeksforgeeks.org/computer-networks/information-assurance-vs-information-security/>

CLASSROOM POLICY	
FACE-TO-FACE DELIVERY	FLEXIBLE TEACHING AND LEARNING ACTIVITIES (FLTAs)
Please refer to the FLTA policies, except for the stipulation on Attendance. Attendance will be checked during face-to-face sessions.	Students’ Responsibilities Students are expected to: a. Take time to read and watch the provided class learning materials. b. Fulfill class requirements such as posting on discussion boards and submitting group presentations. c. Check the status of their submissions and inform the instructor if they are having problems submitting the requirements.



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The instructor may not be able to remind all students about their missing/unsubmitted requirements. If at the end of the course students still have missing requirements, the instructor will compute for the final grade without those submissions (They will be graded with 0) and submit the grade. If the final grade is "INC" grade, the student must contact the instructor to decide in completing the course, ASAP.

d. Confer with classmates or the instructor about the discussions during the synchronous sessions that they missed if unable to attend the live sessions. Recordings will be provided but may not be available immediately.

e. Inform the instructor of concerns that may prevent them from fully participating in the class. This includes withdrawal from the course, long term illnesses, personal matters, and others. Teachers will not be able to help you if you don't tell them of these concerns.

Attendance.
We recognize that students have varying access levels to technology and connectivity, which is why for the most part, the course is conducted asynchronously. During synchronous sessions, there will be no checking of attendance. However, everyone is encouraged to participate in these sessions. A recording and/or transcript of synchronous classes will be made available for students who are unable to attend a synchronous session.

Late Submissions.
Deadlines will be set for the submission of course requirements. This allows the instructor sufficient time to review them. It also helps students pace their work. If you experience problems submitting your work on time, please contact the instructor ahead of time. All requirements must be completed to pass the course. Late submissions will not merit deductions but may not receive timely feedback.

Cheating and Plagiarism.
Any work submitted by a student in this course for academic credit will be the student's own work. Any form of academic dishonesty will be fully dealt with according to the stipulations of the PUP Student Handbook.

Participation.
Participation during discussions is greatly encouraged, and students should feel free to ask questions when an idea or lesson is unclear or confusing, especially during synchronous sessions.

Announcements.
Class announcements will be communicated through the class president. Other communication shall always be coursed through university-issued (i.e., official) email accounts. Students must use their PUP student accounts.

Intellectual Property and Copyright.
Original materials created for this online class is covered by university policies on intellectual property. Some (third-party) instructional materials (e.g., YouTube videos) might be under the public domain or are open access, while others (e.g., some assigned readings) might be subject to fair use. All third-party instructional materials will



	be attributed to their original creators. Electronic copies of the assigned readings are for your private consumption only. Access to the course materials for this class for PUP students will be restricted to only those students registered in this course and will be available for only a
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Mid Term		Final Term	
Class Standing	70%	Class Standing	70%
Exercises/Activities		Exercises/Activities	
Summative Quizzes		Summative Quizzes	
Recitation and Attendance		Recitation and Attendance	
Reflection Paper		Reflection Paper	
Midterm Examinations	30%	Final Examinations	
	100%		30%
			100%

	limited period. Please do not share these materials with anyone elsewhere. The same applies to the recordings of the synchronous sessions.		
COURSE REQUIREMENT/S	GRADING SYSTEM		
Students must develop a mini Information Security Management System (ISMS) for a simulated organization. The project should include policy creation, risk assessment, asset inventory, and control implementation following ISO 27001 standards. This activity integrates all major concepts learned throughout the semester.	Final Grade = (Midterm Grade + Final Term Grade) / 2		



- Compilation of all **hands-on activities**, including asset identification, access control configuration, cryptography practice, and incident management drills. Serves as evidence of individual technical competency and comprehension of key ISO 27001 controls.

CLASS INFORMATION		FACULTY INFORMATION	
Course, Year and Section: BSIT 3-1		Name of Faculty:	
Day and Time:		Consultation Time:	
Room:		Office Tel. No./ Mobile Phone No.	
School Year:	Semester: 2	E-Mail Address:	

Prepared by:	Reviewed and checked by:	Recommending Approval:	Approved by:
Faculty Member	MS. DIANNE S. SEGURORA Head of Academic Programs	DR. RHYAN V. MOLINAR Campus Director	DR. ARMANDO A. TORRES Vice President for Campuses



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